

# Passage

Passage republishes the Sparkplug namespace SparkBridge Host holds as a live OPC-UA server address space, so any OPC-UA client can browse and subscribe without polling the plant.

Free module Ignition 8.1.19+ and 8.3 Runs on: Host

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## 1. Overview

What Passage does

Passage republishes the Sparkplug namespace that SparkBridge Host holds as a live OPC-UA server address space. A non-Sparkplug client (a historian, a SCADA package, an MES, an analytics tool, a test rig) connects to Passage as an ordinary OPC-UA server and browses the same tag tree Host already built, without polling the plant. Passage is not a device driver and it does not write to the field; it exposes what Host already holds.

Passage ships as a free module included with any SparkBridge Edge or Host license.

## 2. What You Need

- A SparkBridge Host gateway (Passage reads Host's tag tree).
- Ignition 8.1.19 or newer, or Ignition 8.3.
- A free TCP port for the OPC-UA endpoint (default 62544).
- An OPC-UA client to consume it: a historian, SCADA package, MES, analytics tool or test rig.

## 3. Install

1. Download the SparkBridge release bundle for your Ignition line from [the download page](#).
2. In the gateway, go to Config > Modules > Install or Upgrade a Module, and upload Provider, then Host, then Edge, in that order.
3. Accept the certificate and license once; confirm all modules show Running. Passage is included, not a separate module upload.
4. Upgrade every installed module together, from the same release bundle, in the same maintenance window.

## 4. License

Passage is free. It is included with any SparkBridge Edge or Host license and ships with no license tag required of its own.

The suite key file, when present, lives at `<data>/sparkbridge-license.key` and is re-checked every 30 seconds, so installing or replacing it takes effect with no restart. From release 2.7.0, SparkBridge modules run on Ignition's two-hour module trial until a verified key names the product; Passage's own license state is not gated the same way since it carries no separate license tag.

## 5. Configure

Current Passage configuration is file/property-based. Set these as JVM system properties in the gateway startup configuration, then restart the gateway.

|                                                               |                                                                                                                     |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>unsbridge.server.enabled</code>                         | Enable the OPC-UA server. Default <code>true</code> .                                                               |
| <code>unsbridge.server.port</code>                            | OPC-UA bind port. Default 62544.                                                                                    |
| <code>unsbridge.server.bindAddress</code>                     | OPC-UA bind address. Default <code>127.0.0.1</code> .                                                               |
| <code>unsbridge.server.hostname</code>                        | Hostname advertised in OPC-UA endpoints. Default <code>localhost</code> .                                           |
| <code>unsbridge.server.path</code>                            | OPC-UA endpoint path. Default <code>/unsbridge</code> .                                                             |
| <code>unsbridge.server.securityDir</code>                     | OPC-UA security directory. Default <code>data/unsbridge/security</code> .                                           |
| <code>unsbridge.spike.enabled</code>                          | Enable the synthetic proof node. Default <code>false</code> .                                                       |
| <code>unsbridge.server.allowAnonymousUnencryptedRemote</code> | Explicit acknowledgement for remote anonymous unencrypted access. Default <code>false</code> .                      |
| <code>unsbridge.birthcache.enabled</code>                     | Persist and replay the last known BIRTH set. Default <code>true</code> .                                            |
| <code>unsbridge.birthcache.file</code>                        | Birth cache path. Default <code>data/unsbridge-birthcache.txt</code> .                                              |
| <code>unsbridge.birthcache.maxAgeHours</code>                 | Maximum accepted cache age in hours. Default 168; 0 disables the age check.                                         |
| <code>unsbridge.birthcache.saveSeconds</code>                 | Cache save interval in seconds. Default 60; minimum 5.                                                              |
| <code>unsbridge.birthcache.deviceGraceSeconds</code>          | Device grace period after an edge NBIRTH. Default 60; minimum 1.                                                    |
| <code>unsbridge.ingest.enabled</code>                         | Enable Sparkplug MQTT ingest. Default <code>false</code> .                                                          |
| <code>unsbridge.broker.uri</code>                             | MQTT broker URI. Default <code>tcp://127.0.0.1:1883</code> .                                                        |
| <code>unsbridge.broker.username</code>                        | MQTT broker username. Default is empty.                                                                             |
| <code>unsbridge.broker.clientid</code>                        | MQTT client id. Default <code>sparkbridge-unsbridge</code> ; use a value beginning with <code>sparkbridge-</code> . |

Store `unsbridge.status.token` and `unsbridge.broker.password` in the file named by `unsbridge.secrets.file`, default `data/unsbridge-secrets.properties`. Passage also supports the shared `sparkbridge.broker.password` fallback for broker access.

## 6. Operate

Passage runs passively: it serves subscriptions and browse requests against Host's live tag tree. It polls nothing in the plant. On an edge NDEATH or DDEATH, the corresponding OPC-UA nodes go to bad quality within the configured device grace period. On gateway restart, the birth cache (when enabled) replays the last known BIRTH set so clients see a populated address space immediately rather than an empty tree while Host re-establishes its own sessions.

## 7. Verify

Connect a third-party OPC-UA client (for example UaExpert) to the configured endpoint, browse the address space, and subscribe to a value. Confirm the value matches the Sparkplug wire within one update, with the source timestamp preserved. Confirm a node goes to bad quality within the death window after an NDEATH/DDEATH.

## 8. Troubleshoot

| Symptom                                                   | Cause                                                                              | Fix                                                                                                     |
|-----------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Client cannot connect                                     | <code>unsbridge.server.enabled</code> is false, or the port is blocked             | Confirm the property is <code>true</code> and the configured port is reachable from the client          |
| Remote client refused with an anonymous/unencrypted error | <code>unsbridge.server.allowAnonymousUnencryptedRemote</code> is false (default)   | Set it to <code>true</code> only after accepting the security tradeoff, or configure a secured endpoint |
| Address space empty after restart                         | Birth cache disabled or stale beyond <code>unsbridge.birthcache.maxAgeHours</code> | Enable the birth cache, or wait for Host to re-establish live BIRTH data                                |
| Broker-backed ingest not updating                         | <code>unsbridge.ingest.enabled</code> is false (default)                           | Set it to <code>true</code> and confirm <code>unsbridge.broker.uri</code> and credentials               |

## 9. Reference

- Default OPC-UA port: 62544.
- Default endpoint path: `/unsbridge`.
- Secrets file default: `data/unsbridge-secrets.properties`.
- Birth cache file default: `data/unsbridge-birthcache.txt`.

## 10. Support

[Passage in the docs](#) · [Contact](#) · [GitHub releases](#)